JUNIA Grande école d'ingénieurs

HEI·ISEN·ISA

Septembre -Décembre 2025

ISEN
Module
Numérique Responsable

Courses objective & planning

Teachers

- Brice PASIAN – Groupes 1 & 2
- Olivia BERTOUT – Apprenti, Groupes 3 & 4
- Contributors : Christophe Denoyelle, INR, Thomy Gensollen, MC2i, Atelier 21, Latitudes

Brice Pasian



- **2007 / 2020** : Developer > Project Leader > Service Center Director
- **Since 2021** : Independent consultant in Responsible Digital *helping organizations to reduce the environmental and social footprint of digital technology.*

- *Publications (as author or contributor)*

- «Environmental impacts of digital technology in the world, third edition», Green IT Association, 2025 (IENM 2025), https://t.ly/greeniteco.IENM2025

- Web Eco-design: 115 best practices - 4th edition, 2022

Objectives

- Understand environmental and social challenges of digital technology
- Master the fundamentals of the Responsible Digital approach and its best practices, through the concepts of:
  - Environmental assessment including life cycle analysis
  - Eco-design of digital services
  - Digital accessibility
- Have insight from professionals in the field

Lessons - 10 hours
MOOC - 6 hours
Workshop/Tutorials - 6 hours
Insights - 6 hours

Planning

Groupe 1

DATE	HOUR	CONTENT
September 17	8.00	Main environmental and social challenges of digital Digital Responsible
	10.20	MOOC INR – Modules 1 - 3
September 24	8.00	The components of the Information System - Zoom / Datacenter & IA
	10.20	MOOC INR – Modules 4 - 11
October 1st	8.00	AI Battle (Association Latitudes)
	10.20	TP Devices diagnosis (Christophe Denoyelle)
October 8	8.00	Environmental assessment / Part 1: Life Cycle Assessment (LCA)
	10.20	MOOC INR – Modules 12 à 14 – Certification preparation
November 12	8.00	Environmental assessment / Part 2: LCA in relation to TP Diagnosis Digital accessibility
	10.20	Testimonial Alumni ISEN – Thomy Gensollen
November 19	10.20	Testimonial MC2i "Making digital transformation a positive revolution"
	13.30	Eco-design of digital services - TD eco design
November 26	8.00	TD presentation (evaluation) - Conclusion
	10.20	Testimonial Atelier 21 – Low Tech

Planning

Groupe 2

DATE	HOUR	CONTENT
September 17	8.00	MOOC INR – Modules 1 à 3
	10.20	Enjeux environnementaux et sociaux du numérique Numérique Responsable
September 24	8.00	MOOC INR – Modules 4 à 11
	10.20	Les composants du Système d'information - Zoom / Datacenter & IA
October 1st	8.00	TP Diagnostic équipements (Christophe Denoyelle)
	10.20	Bataille de l'IA (Association Latitudes)
October 8	8.00	MOOC INR – Modules 12 à 14 – Préparation Certification
	10.20	L'évaluation environnementale / Partie 1 : L'Analyse du Cycle de Vie (ACV)
November 12	8.00	Témoignage Alumni ISEN – Thomy Gensollen
	10.20	L'évaluation environnementale / Fin de vie et lien avec TP Diagnostic L'accessibilité numérique
November 19	8.00	Témoignage MC2i *"Faire de la transformation numérique une révolution positive"*
	15.50	Eco conception de services numériques - TD écoconception
November 26	8.00	Intervention Atelier 21 – Low Tech
	10.20	Restitution TD (évaluation) - Conclusion et ouverture

Evaluation

Continuous assessment | 30%

- 40% - MOOC INR certificate ("Institut Numérique Responsable")
- 60% - TD Eco-design of digital services

Final Evaluation| 70%

- Questions about the entire courses

>> Wed. 3rd Dec.



Evaluation – MOOC INR

MOOC INR

- https://www.academie-nr.org/

Evaluation

- Duration : 40'
- 40 questions : min. Score of 72,5% for validation
- /!\ No document, No double screen

Required equipment:

- Identification Card
- Webcam
- Room where you can isolate yourself (to be verified via webcam) for the entire duration of the evaluation

Access open from October 6.

Additional questions ? Please contact yohan.giraudeau@junia.com



Let's start !

Lesson 1 - Today

- Uses of Digital Technology
- Environmental impacts of digital technology
- Social impacts of digital technology
- Digital Responsible



How many smartphones do you have ?

(in use or not)

1

2

3

4+



What is the main source of environmental impact of digital technology?

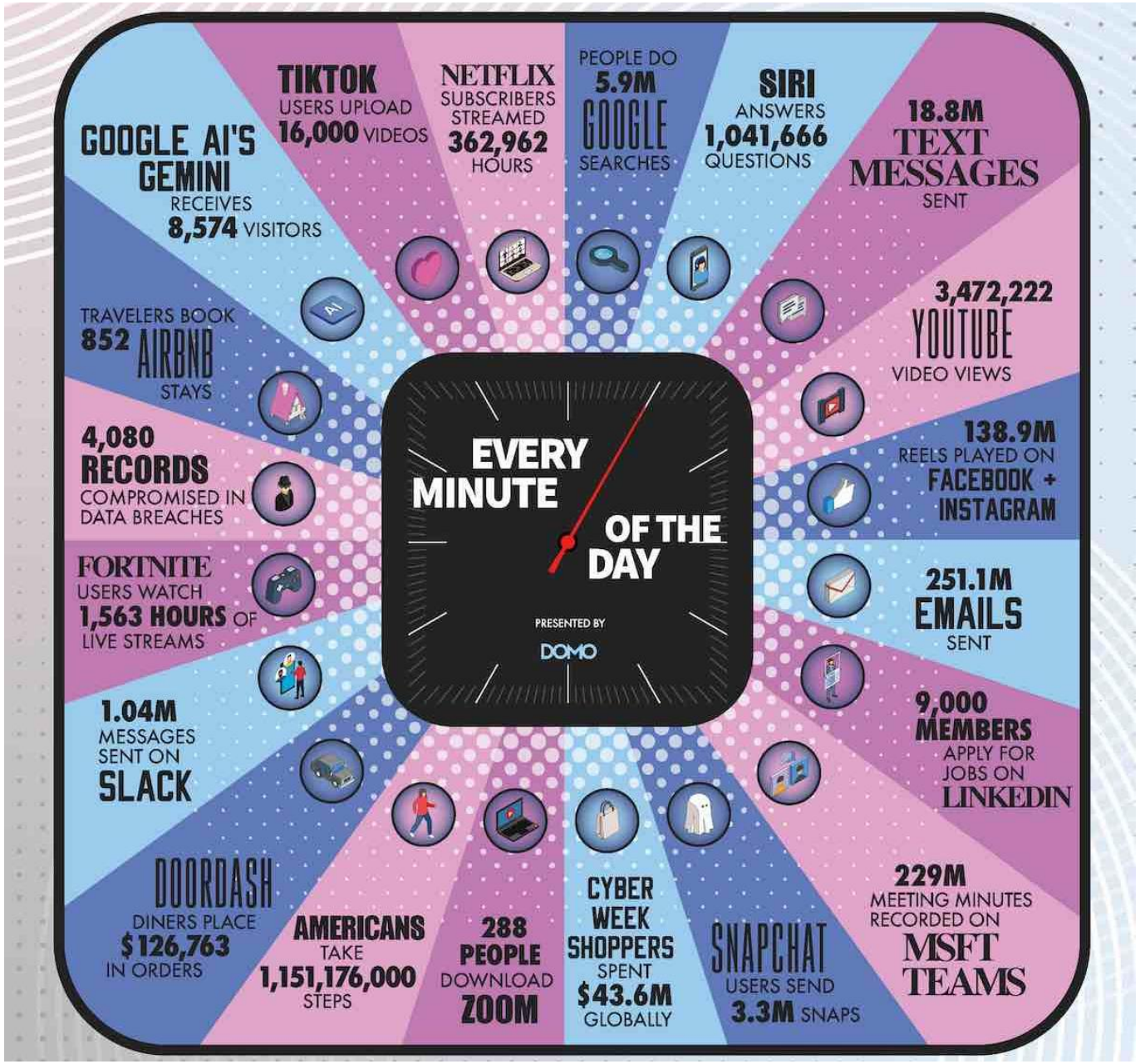
1. Data center energy consumption
2. Manufacturing of user equipments
3. Energy consumption of user equipment



What happens in 1 minute on Internet ?

251 millions		Netflix streamed hours
362 962		Tiktok users video uploads
16 000		Youtube video views
3,4 millions		Emails Sent

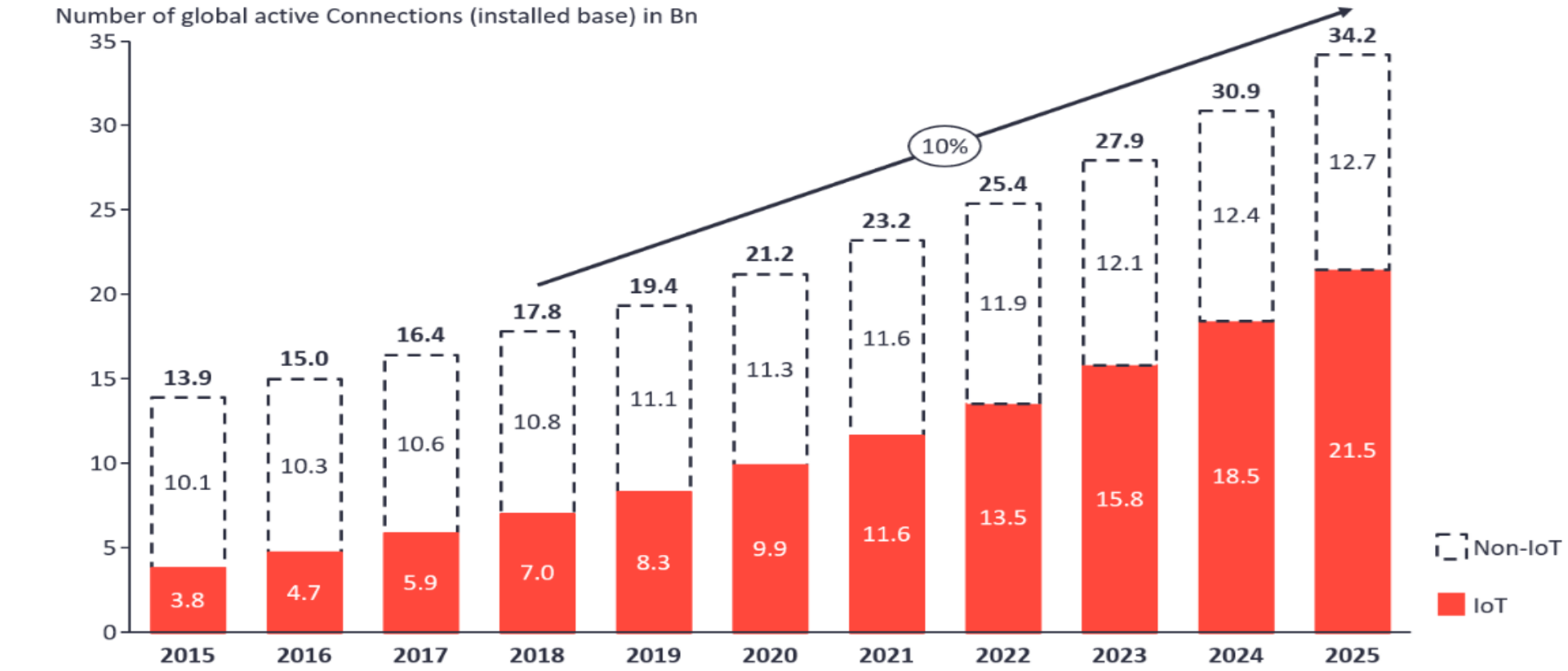
Every Minute on Internet in 2024



https://www.blogdumoderateur.com/infographie-une-minute-internet-2024/

And the forecasts are….

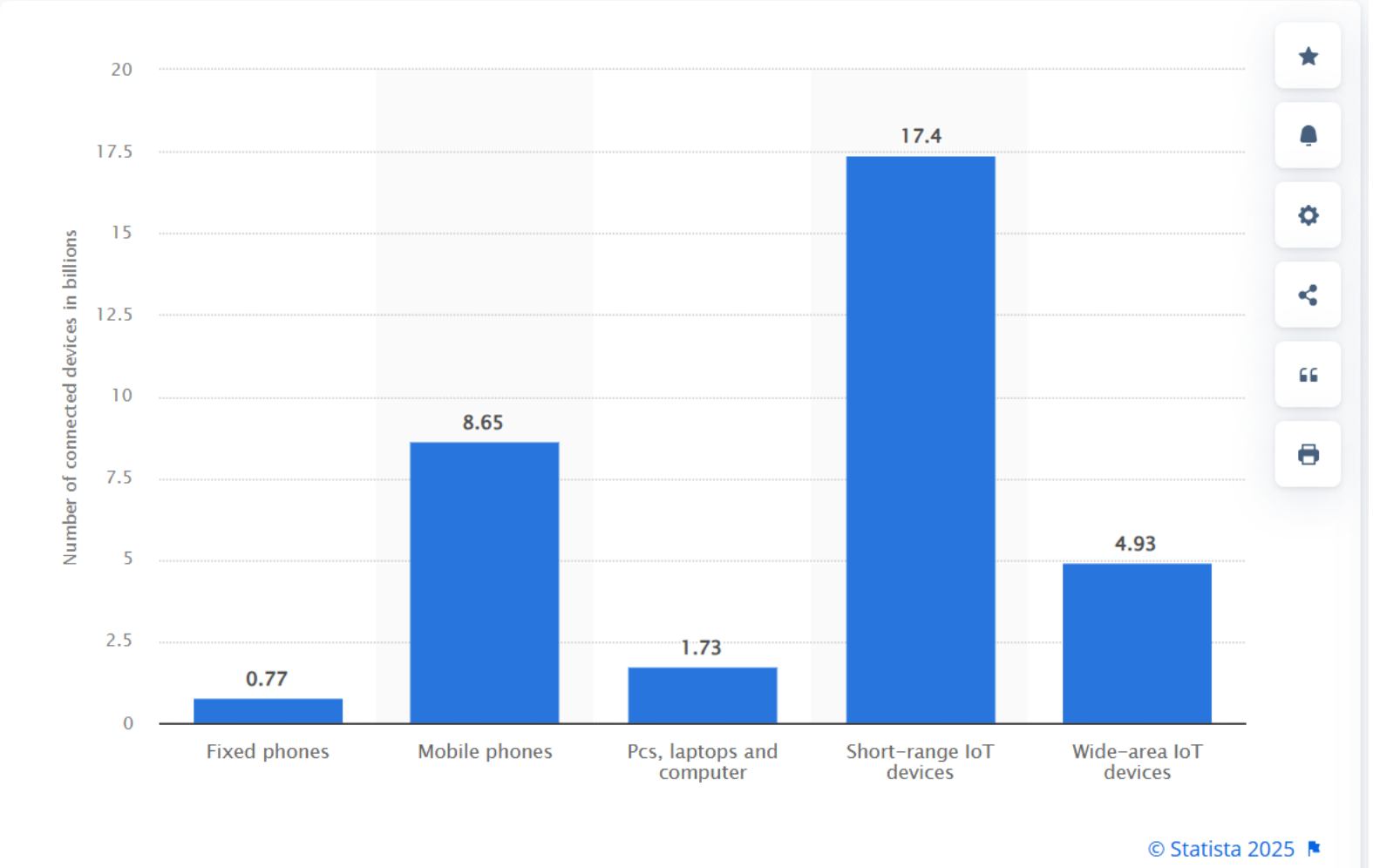
Total number of active device connections worldwide



Note: Non-IoT includes all mobile phones, tablets, PCs, laptops, and fixed line phones. IoT includes all consumer and B2B devices connected – see IoT break-down for further details
Source: IoT Analytics Research 2018

Number of connected devices worldwide as of February 2025, by device

Non IoT - 11
IoT - 22, 1



https://www.statista.com/statistics/1559435/connected-devices-worldwide/

And you,
when was the last time you went out without your phone ?



New uses

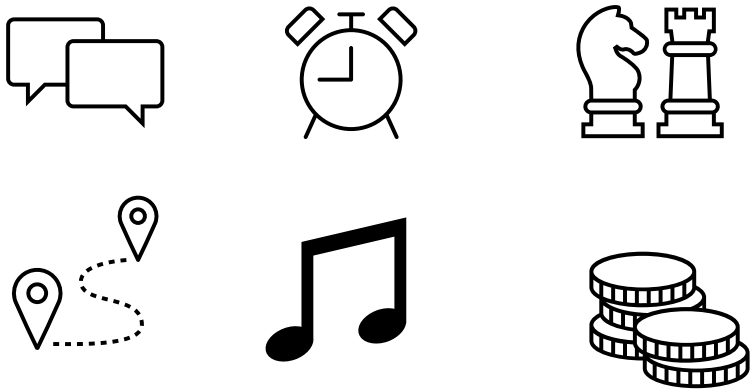
- Can you give examples of digital uses that **you didn't have** 5 years ago?

- What do you think about **impacts** ?



New uses

- What is the mobile app you can't live without?







72% of French people consider important to have permanent access to the internet

TRUE / FALSE



72% of French people consider important to have permanent access to the internet

TRUE / FALSE

Source : https://www.greenit.fr/2021/07/06/63-des-francais-accros-au-numerique/

And to answer our uses, we need to produce equipments, a lot of equipments….



+31 billion

of phones then smartphones sold between 2000 et 2022

Source : https://www.gsma.com/solutions-and-impact/connectivity-for-good/external-affairs/wp-content/uploads/2023/06/Research-Methodology-2023.pdf

And the forecasts are….

Table 1. Mobile subscribers as a percentage of regional population

Region	2018	2023
Global	66%	71%
Asia Pacific	65%	72%
Central and Eastern Europe	79%	81%
Latin America	75%	78%
Middle East and Africa	53%	57%
North America	86%	88%
Western Europe	84%	85%

Source: *Cisco Annual Internet Report*

Table 2. Network generation distribution

	3G and Below	4G	5G	LPWA*
% share				
Global				
Global	29%	46%	11%	14%
By region				
North America	1%	45%	17%	37%
Western Europe	13%	43%	16%	28%
Central and Eastern Europe	31%	50%	2%	16%
Asia Pacific	23%	52%	13%	12%
Latin America	37%	50%	7%	6%
Middle East and Africa	73%	22%	1%	4%

* Low-Power Wide-Area (LPWA) is an ultra-narrowband wireless network meant specifically for M2M modules that require low bandwidth and wide geographic coverage.

Source: *Cisco Annual Internet Report*

By 2023, **71%** of the global population will have a mobile subscription.

https://www.cisco.com/c/dam/en/us/solutions/collateral/executive-perspectives/annual-internet-report/air-executive-summary-pgr-mobility.pdf

ENVIRONNEMENTAL impacts of digital technology

The digital "world"



Equipments **+30 billion**
Users +5,3 billion
6 equipment/user



Equipments 651 million
Users 58 million
11 equipment/French people



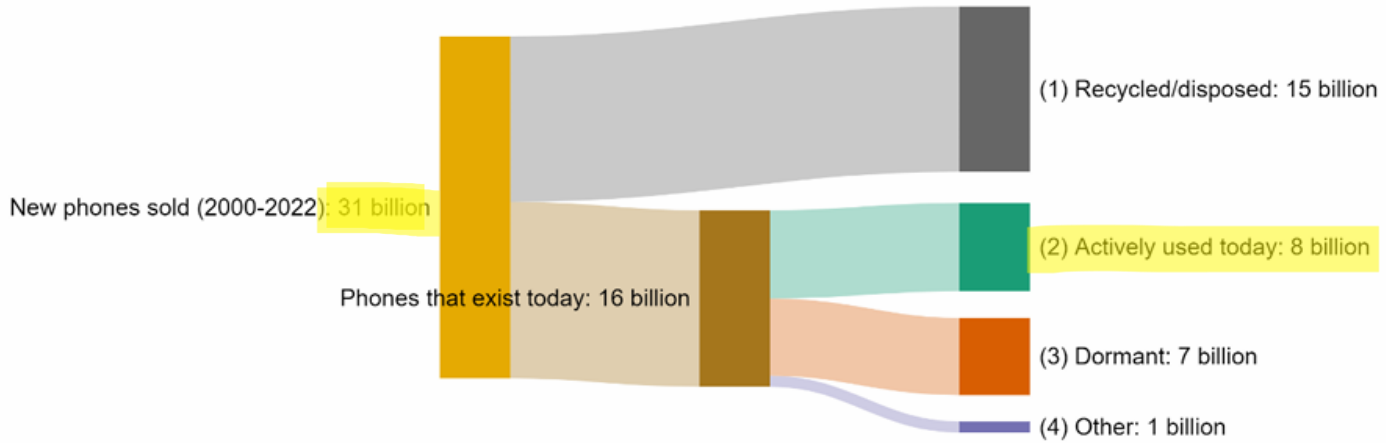
Source : « *GreenIT.fr - Impacts environnementaux du numérique dans le monde 2025*
et étude « iNUM : impacts environnementaux du numérique en France », GreenIT.fr, 2020, chiffres mis à jour.

The digital "world"

In use Digital equipments **+30 billion**



Figure 1 – Summary of top-down estimate based on sales, stock, and connections data



Notes: "Other" includes used phones not included in the remaining three categories, e.g. used phones that are in the process of being resold on online marketplaces and second-hand shops.

Sources: New phones sold (GSMA analysis based on Gartner, IDC, and other sources); phones that exist today (UNITAR, 2022 based on Forti et al., 2020); phones actively used today (GSMA Intelligence, 2023); recycled/disposed (calculated based on the number of possessed phones estimated by UNITAR, 2022).

Source : « *GreenIT.fr - Impacts environnementaux du numérique dans le monde 2025*
et étude « iNUM : impacts environnementaux du numérique en France », GreenIT.fr, 2020, chiffres mis à jour.
Source : https://www.gsma.com/solutions-and-impact/connectivity-for-good/external-affairs/wp-content/uploads/2023/06/Research-Methodology-2023.pdf

Only one planet

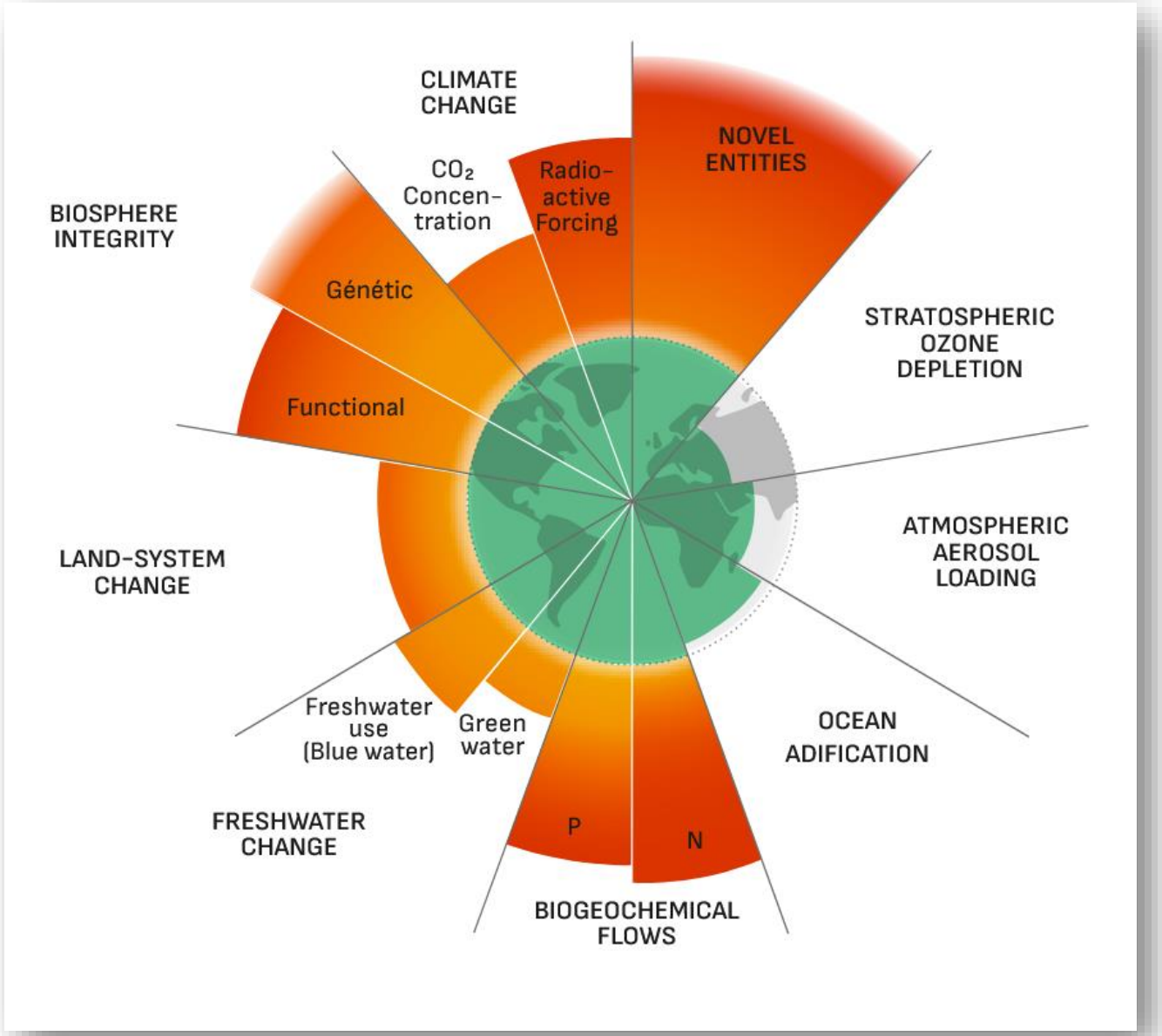


Planetary boundaries

Digital technology impacts negatively a lots of them...

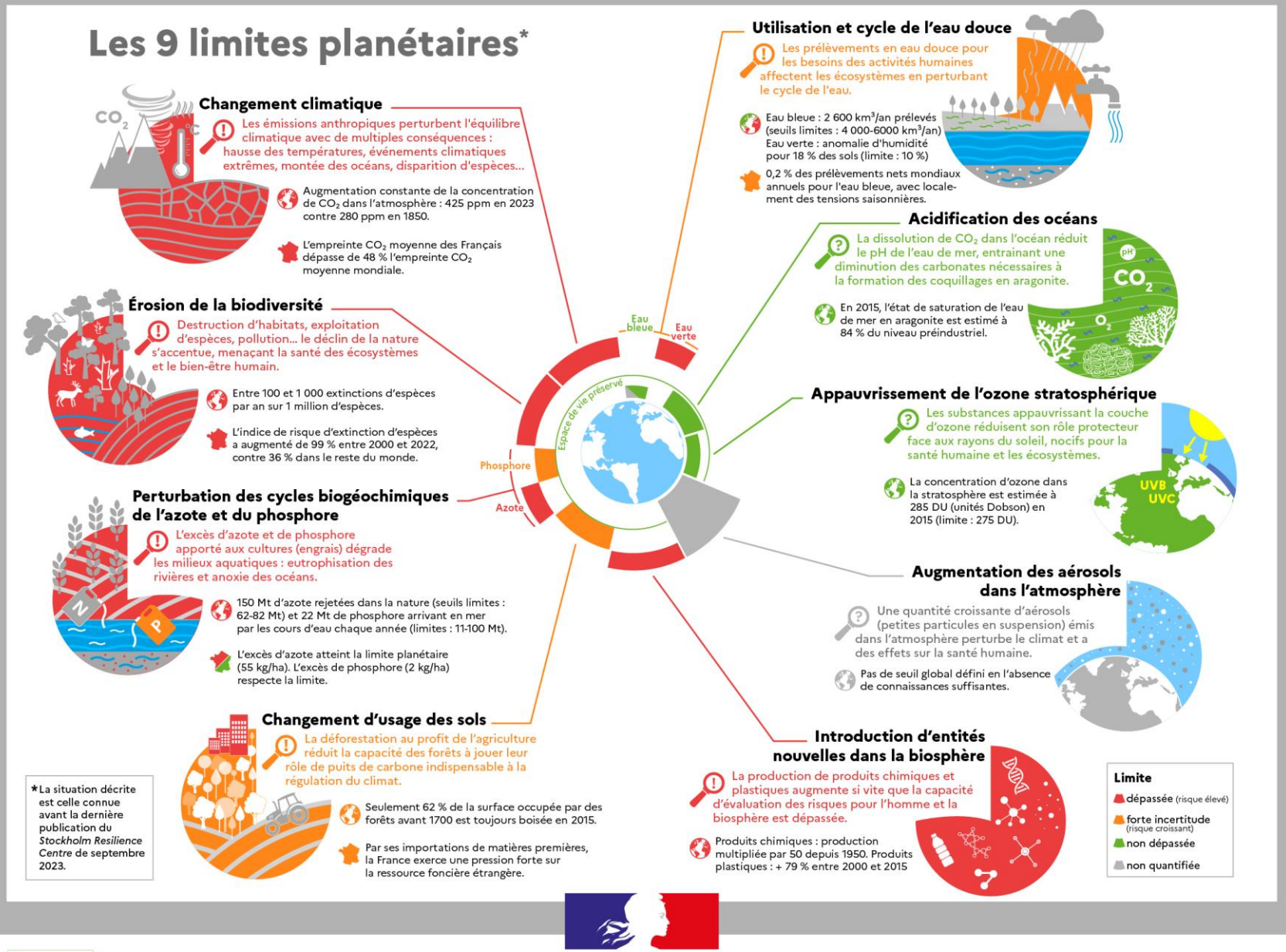
- Climate change
- Erosion of biodiversity
- New entities
- Chemical pollution
- Water cycle

And also, resource depletion, waste and energy use.



Source : Earth beyond six of nine planetary boundaries | Science Advances, https://www.science.org/doi/10.1126/sciadv.adh2458

Les 9 limites planétaires*



*La situation décrite est celle connue avant la dernière publication du *Stockholm Resilience Centre* de septembre 2023.

© CGDD/SDES, 2023 - Bertrand Gaillet



La France face aux neuf limites planétaires

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Distribution of digital impacts

386:1

	30,5 billion		79 million
			
	User equipments	Network	Datacenters
 Global warming potential (kg CO2 eq)	54 %	23 %	23 %
 Depletion of abiotic resources, “materials” (kg Sb eq)	72 %	15 %	13 %
 Depletion of abiotic, “fossil” resources (MJ)	53 %	24 %	22 %

Source : « *GreenIT.fr - Impacts environnementaux du numérique dans le monde 2025*

Exponential growth of digital technology

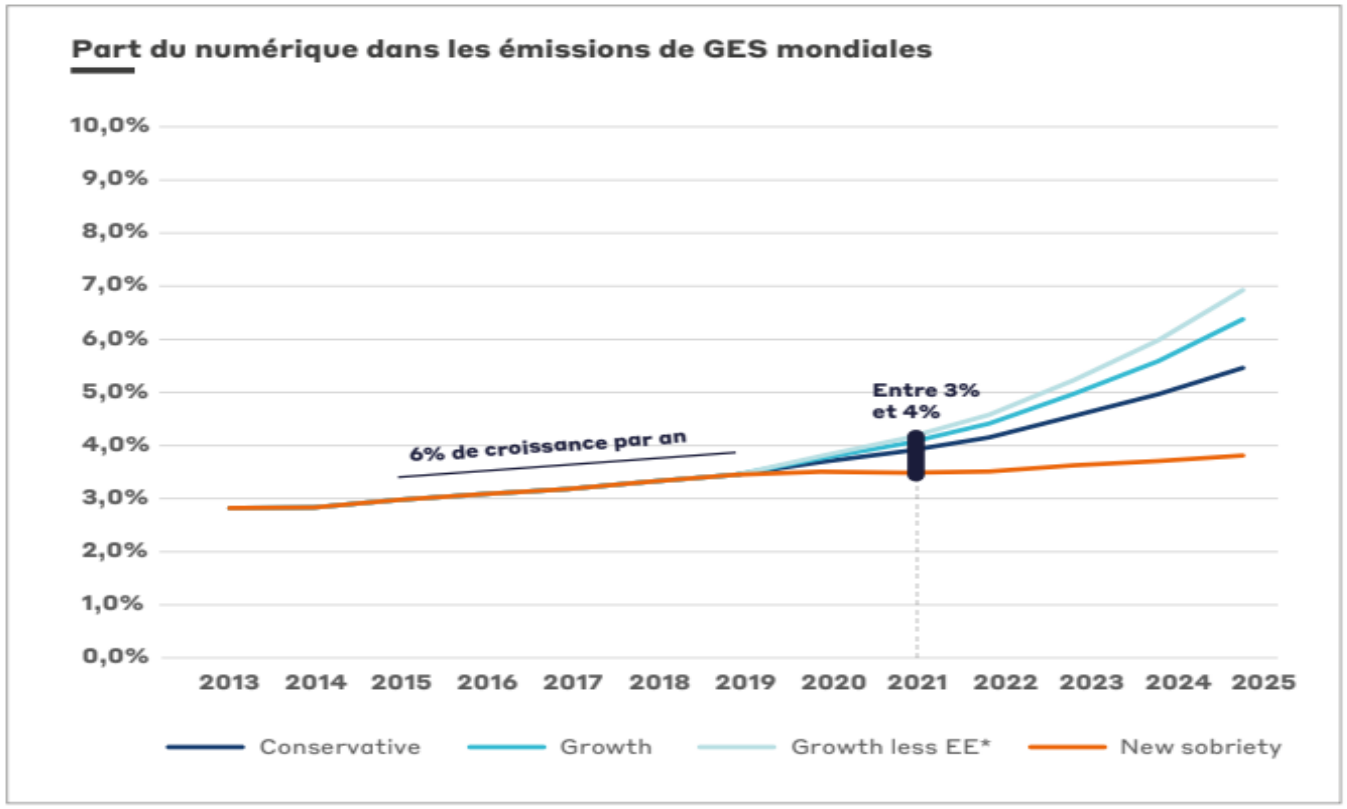
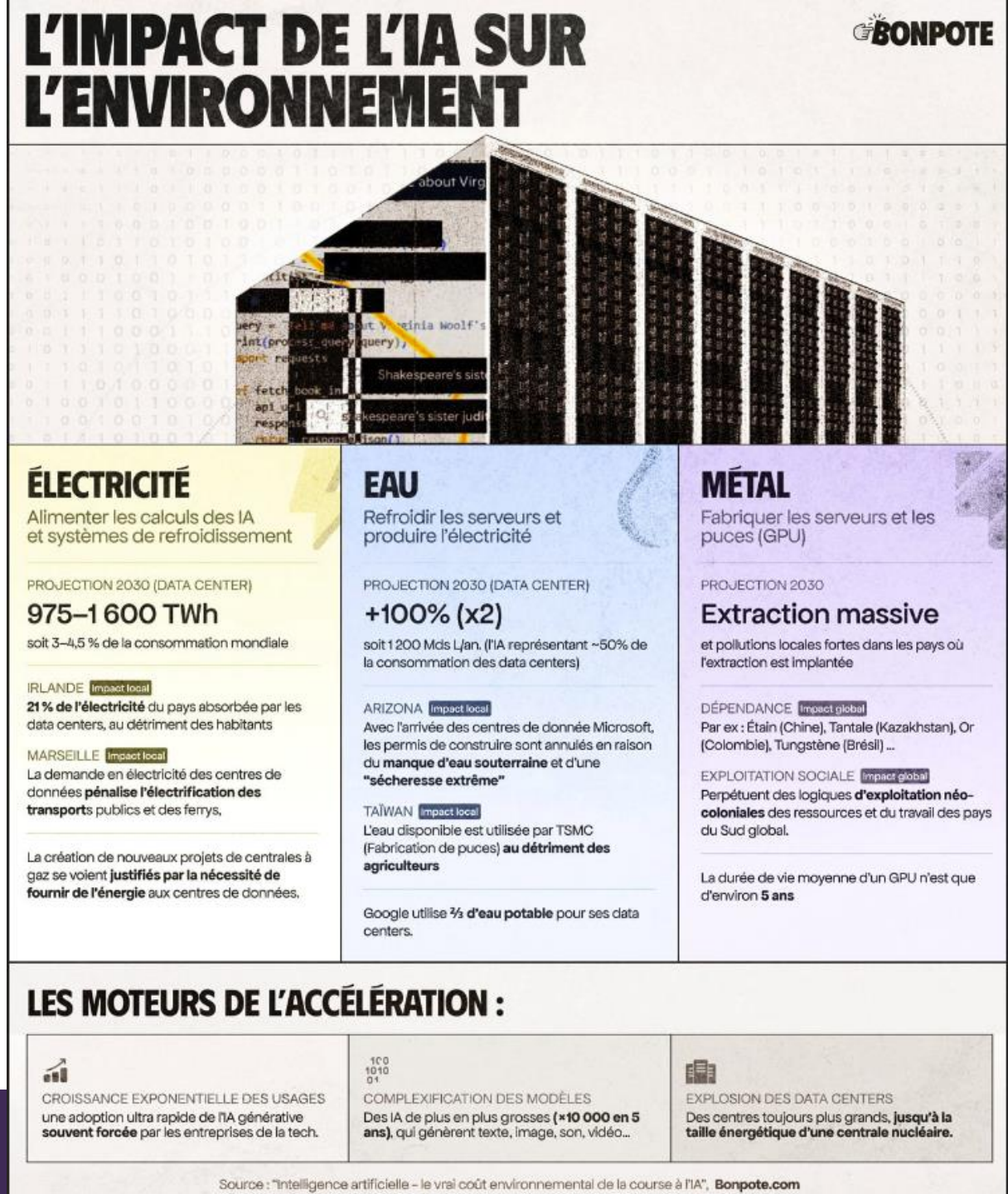


Figure 1 : Évolution 2013-2025 de la part du numérique dans les émissions de GES mondiales *(The Shift Project – Forecast Model 2021)*
** Growth less Energy Efficiency (traduction : croissance avec un ralentissement des gains d'éfficacité énergétique)*

… and exponential use of AI



Annual sustainable budget per digital user

Annual global digital usage consumes
40% of the annual sustainable GHG budget per Internet user
to stay below 1.5°C of global warming.



Global warming potential
(kg CO2 eq)

40 %

Digital part



Depletion of abiotic resources, “materials”
(kg Sb eq)

28 %

Digital part



Fine particles

23 %

Digital part

Source : *GreenIT.fr - "Impacts environnementaux du numérique dans le monde 2025"*

My daily digital footprint









		Primary energy	Global warming potential	Water cycle stress	Depletion of abiotic ressources
French people *(58 million internet users)*	**By day / person**	1000 W Electric radiator during 8 hour	6 kms in combustion car	2,5 packs of mineral water (6l)	197 kg of excavated soil
	For a year / person	11 000 MJ	330 kg CO2 e	5 500 litres	72 tons

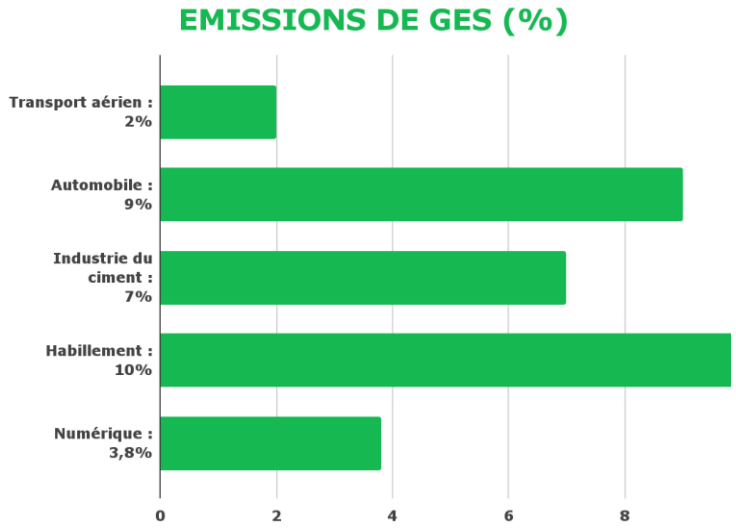
Source : *GreenIT.fr, 2021 « iNum : impacts environnementaux du numérique en France »*

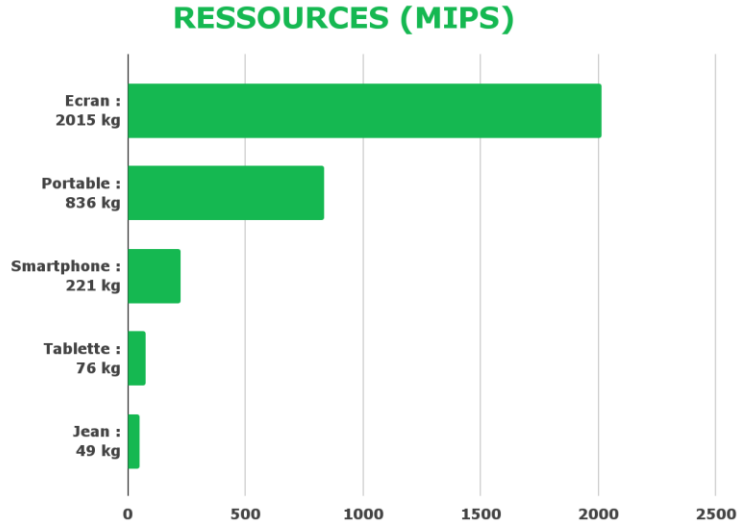
Digital footprint - Comparison

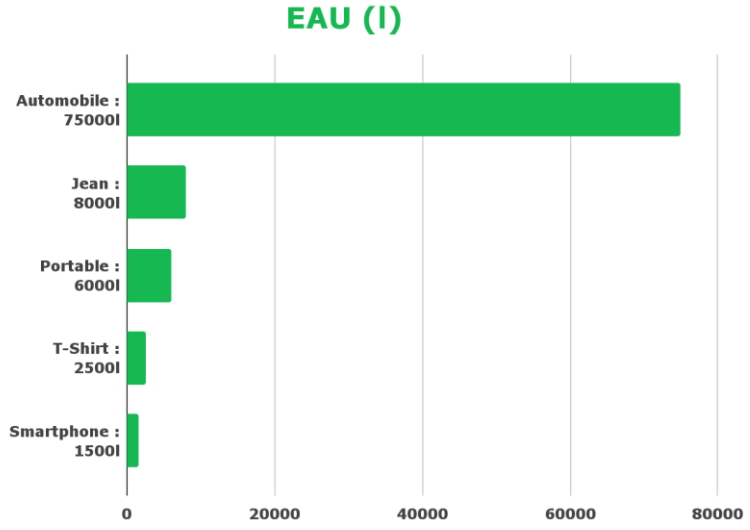






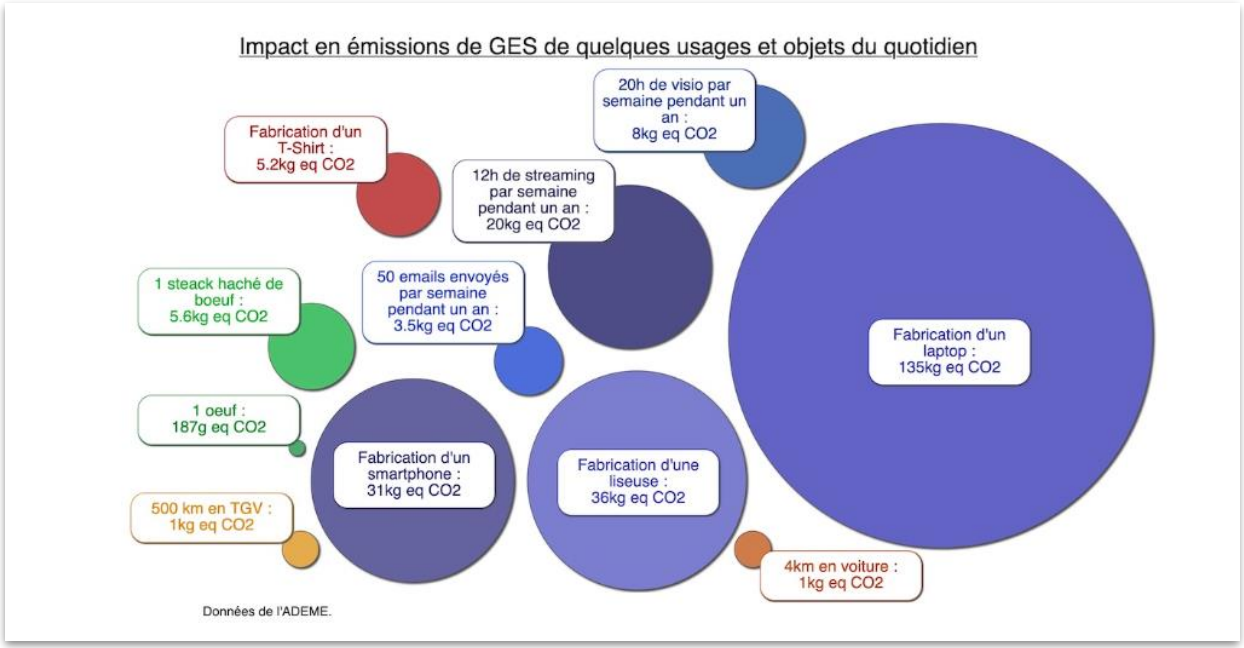






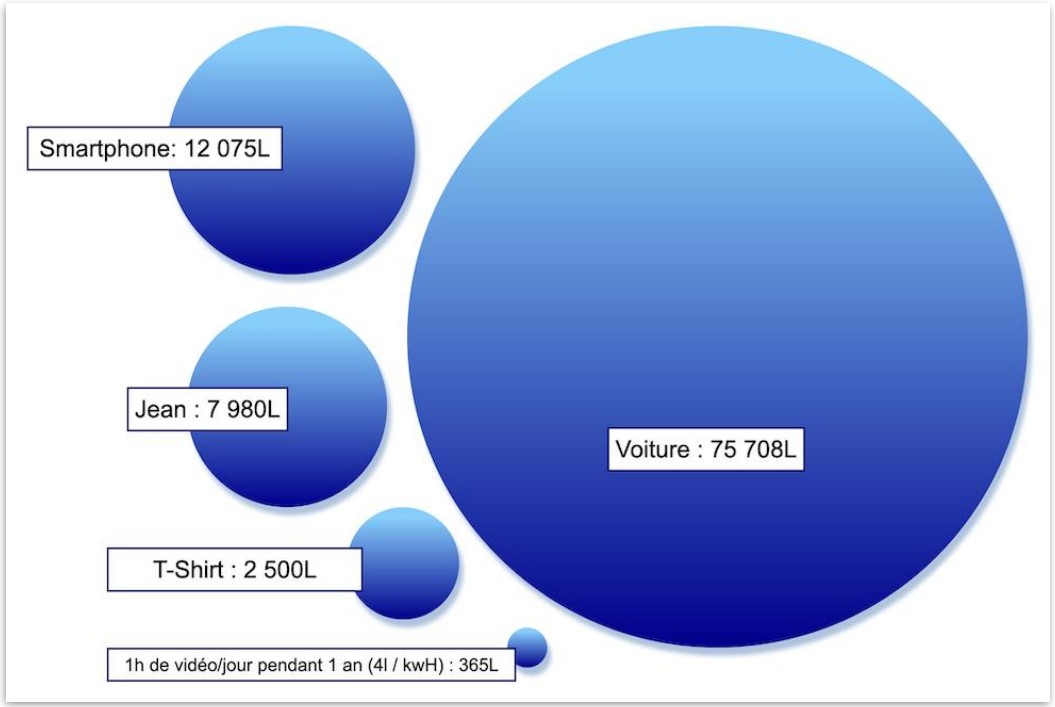
sources : GreenITfr - Sénat - The Shift project

Some equivalences



GREENHOUSE GAS

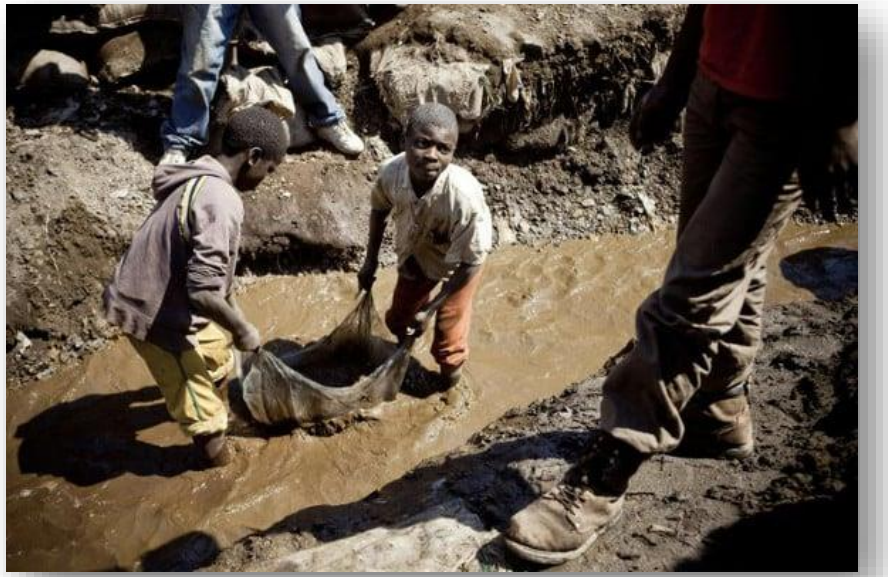
WATER CONSUMPTION



Sources : http://raphael-lemaire.com/2022/12/11/mep-maj/

SOCIAL impacts of digital technology

Social impacts during manufacturing



Forced Labor and Child Labor



Armed conflicts

40,000 Children in the Democratic Republic of Congo

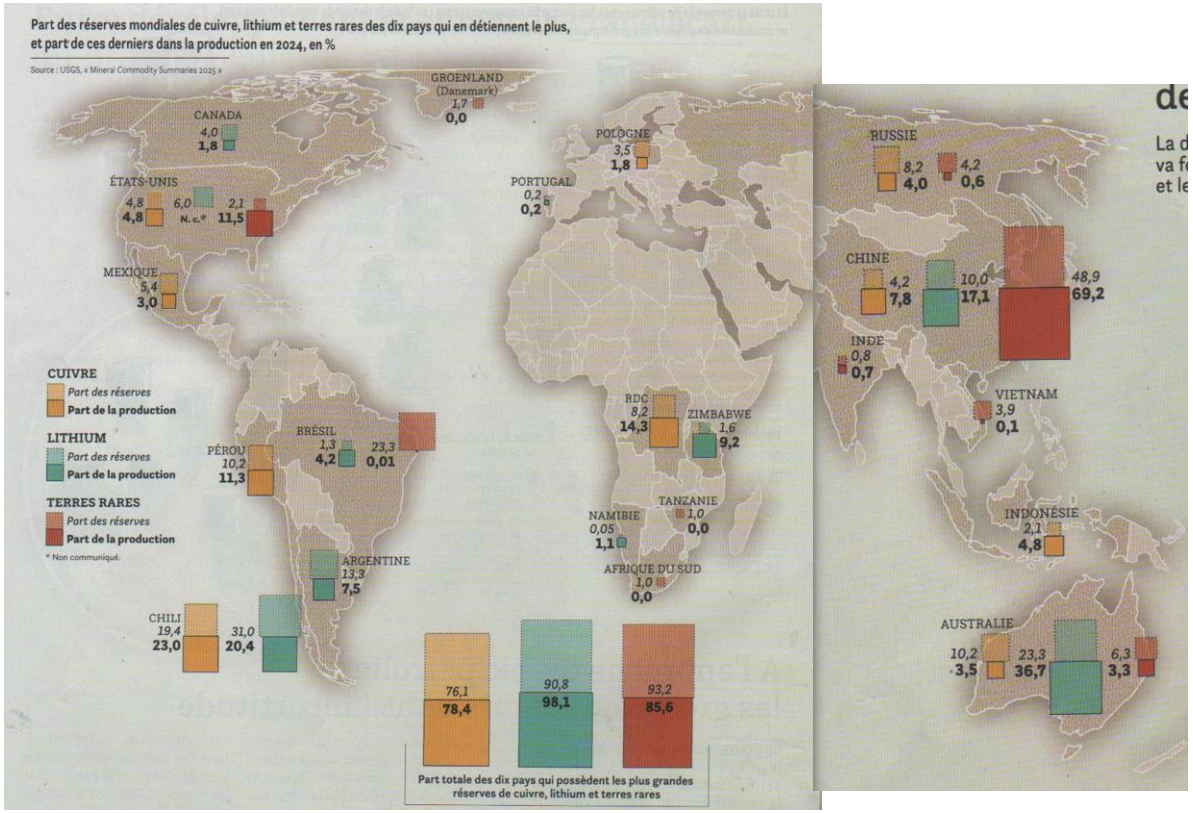
USEFUL LINKS

Mon smartphone est-il lié au travail des enfants ?,

Le travail des enfants derrière la production de smartphones et de voitures électriques,

https://www.globalwitness.org/fr/itsci-laundromat-fr/

https://archives.qqf.fr/infographie/52/smartphone



L'iPhone au cœur de la guerre commerciale

Depuis cinq ans, Apple a diversifié sa chaîne d'approvisionnement, sans toutefois réduire sa dépendance à la Chine. La forte imbrication des chaînes de valeur de ses iPhone l'empêche de rapatrier sa production aux Etats-Unis.

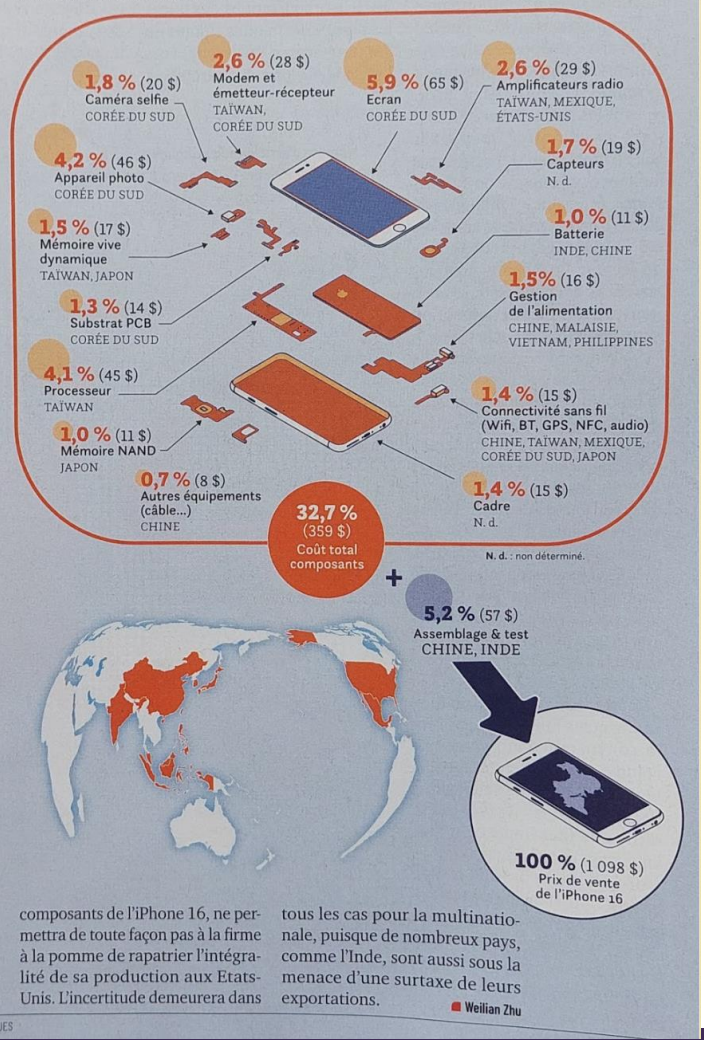
Ces derniers mois, les utilisateurs américains de produits électroniques ont vécu au rythme des annonces de Donald Trump. La mise en place de taxes douanières de 145 % sur les produits en provenance de Chine a d'abord créé un vent de panique : de nombreux produits électroniques sont concernés par cette nouvelle mesure et auraient pu voir leur prix exploser. Puis quelques jours après, dans un revirement dont il a le secret, le président américain annonce finalement l'exemption des produits électroniques de ces surtaxes, maintenant toutefois le taux préexistant de 20 % sur les importations chinoises. Parmi les acteurs concernés par cette incertitude, Apple est particulièrement exposée : l'iPhone, qui représente 50 % de ses revenus, est à 85 % assemblé en Chine.

En 2019, lors du premier mandat de Donald Trump et de sa guerre commerciale, la multinationale avait réussi à négocier des exemptions de taxes sur des composants électroniques importés de Chine. Depuis, elle a cherché à réduire sa dépendance vis-à-vis de cette dernière. En mai dernier, son PDG Tim Cook expliquait à des investisseurs que *« la majorité des iPhone vendus aux Etats-Unis aura l'Inde pour pays d'origine »*, tandis que le Vietnam assemblera *« quasi tous les iPad, Mac, Apple Watch et AirPods vendus aux Etats-Unis »*. Une façon d'anticiper les prochaines tensions entre les Etats-Unis et la Chine sans toutefois réduire, à moyen terme, sa dépendance à cette dernière, le pays restant le principal fabricant d'iPhone destinés au reste du monde.

La forte imbrication des chaînes de valeur d'Apple, illustrée par les lieux de fabrication des composants de l'iPhone 16, ne permettra de toute façon pas à la firme à la pomme de rapatrier l'intégralité de sa production aux Etats-Unis. L'incertitude demeurera dans tous les cas pour la multinationale, puisque de nombreux pays, comme l'Inde, sont aussi sous la menace d'une surtaxe de leurs exportations.

Estimation de la part du prix de chaque composant clé d'un iPhone 16 de 128 Go dans le prix final (en %), prix de chaque composant (en dollars) et pays de fabrication

Sources : appleinsider.com, fr.ifixit.com et TD Cowen



Weilian Zhu

Sources : Alternatives Economiques – Juillet-Aout 2025

Mine in DRC



https://www.youtube.com/watch?v=tbgJf_EkDv4
https://www.youtube.com/watch?v=2Y7W0Bs162o *(1.40-3.00)*

And at the end of the lifecycle



60% of digital waste are part of traffic and does not follow WEEE processing channels

45,000 children

sort and recover components and "precious" metals from electronic waste in Delhi's landfills

USEFUL LINKS

Delhi, capitale mondiale des déchets électroniques,

Decharge d'Agbogbloshie au Ghana

10 sites les plus pollués du monde

Social impacts about uses

Privacy

Ethics

Security

Health

Accessibility

Privacy, Ethics and Dependence

The goal of digital industry companies > Make Money

They sell to advertisers their ability to **influence our behavior and our way of thinking.**

To do this, they need **a lot of data** to predict our next actions because they want to keep us in as long as possible on their application or website, they want to build loyalty...



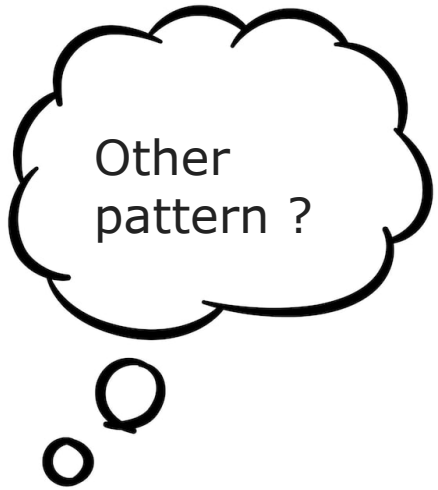
Using dark patterns

Streaming : Video auto-launch

Social Networks : infinite scroll, notification, content renew...

Newsletter: Make unsuscribe so complex...

Booking : /!\Only 1 room left



Pour aller plus loin : **Concevoir sans dark patterns** par les designers ethiques

Example



https://www.youtube.com/watch?v=fx5VZFdkuf4

Privacy & Security

Risks associated with social engineering:

- Respect for privacy (GDPR)
- Stress related to monitoring
- Decrease or loss of the trust about data policies
- Unintentional discrimination and bias
- Censorship

General Data Protection Regulation
GDPR *(fr, RGPD)*

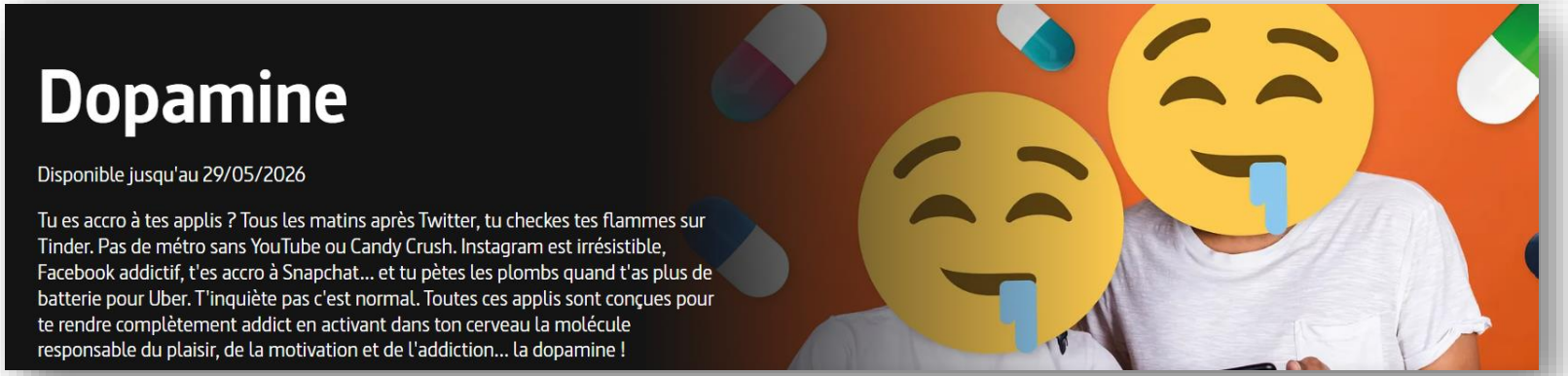
This Regulation establishes **rules on the protection** of physical persons about the processing of personal data and rules on the free movement of such data.

This Regulation **protects the fundamental rights and freedoms** of physical persons, and in particular their right to the protection of personal data.

Pour aller plus loin : **Du mauvais usage de l'ingénierie sociale face aux libertés privées et publiques : du marketing à la cybercriminalité.**

Addiction…





Sources :
https://neurolaunch.com/dopamine-texting/
https://www.arte.tv/fr/videos/RC-017841/dopamine/

Social impacts : IA challenges



A political figure *A secretary*

- **ALGORITHMIC BIAS**

AI tools are trained on data that may contain inherent biases, which can lead to discrimination or stereotypes in their results.

- **CREATIVITY**

AI is now capable of generating content, and intellectual property issues are becoming complex. We face challenges in respecting the property rights of artists and regulating the generation of works by AI.



- **FIABILITY**

How to - now - maintain trust in information ?

- **SOCIETY & PERSONAL DATA**

AI will influence how we “build society”, particularly our relationship to work and education, international relations, and the legal framework.

source : **Association Latitudes**

Another social impact : Digital accessibility



15% of French people do not know how to use digital and internet ?

TRUE / FALSE



15% of French people do not know how to use digital and internet ?

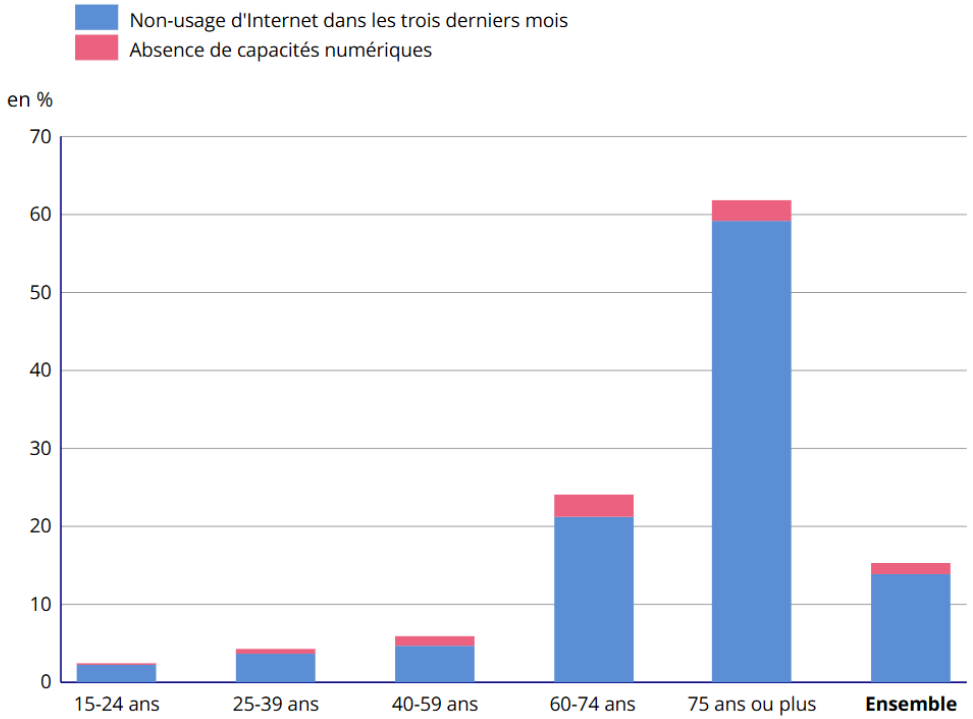
TRUE / FALSE

https://www.anlci.gouv.fr/illectronisme/lillectronisme-en-chiffres/

Digital is not accessible to everyone

Some figures in France

Figure 1 – Taux et décomposition de l'illectronisme selon l'âge

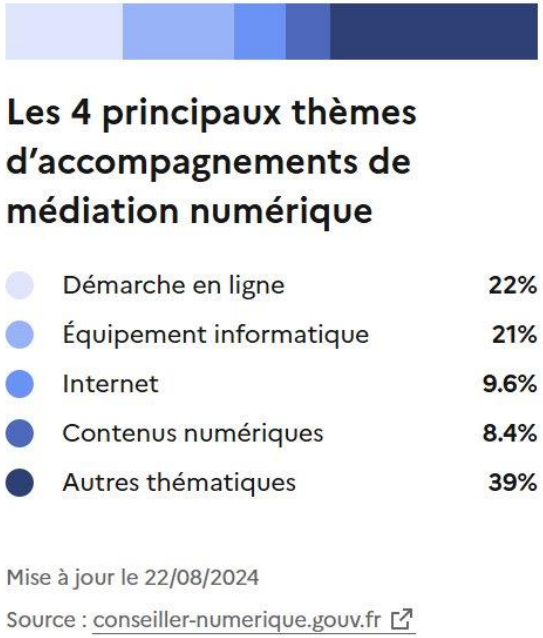


Lecture : en 2021, 59,2 % des personnes de 75 ans ou plus n'ont pas utilisé Internet dans les trois derniers mois et 2,7 % n'ont pas les compétences numériques de base. Au total, 61,9 % sont en situation d'illectronisme.

Champ : France hors Mayotte, personnes de 15 ans ou plus vivant en logement ordinaire.

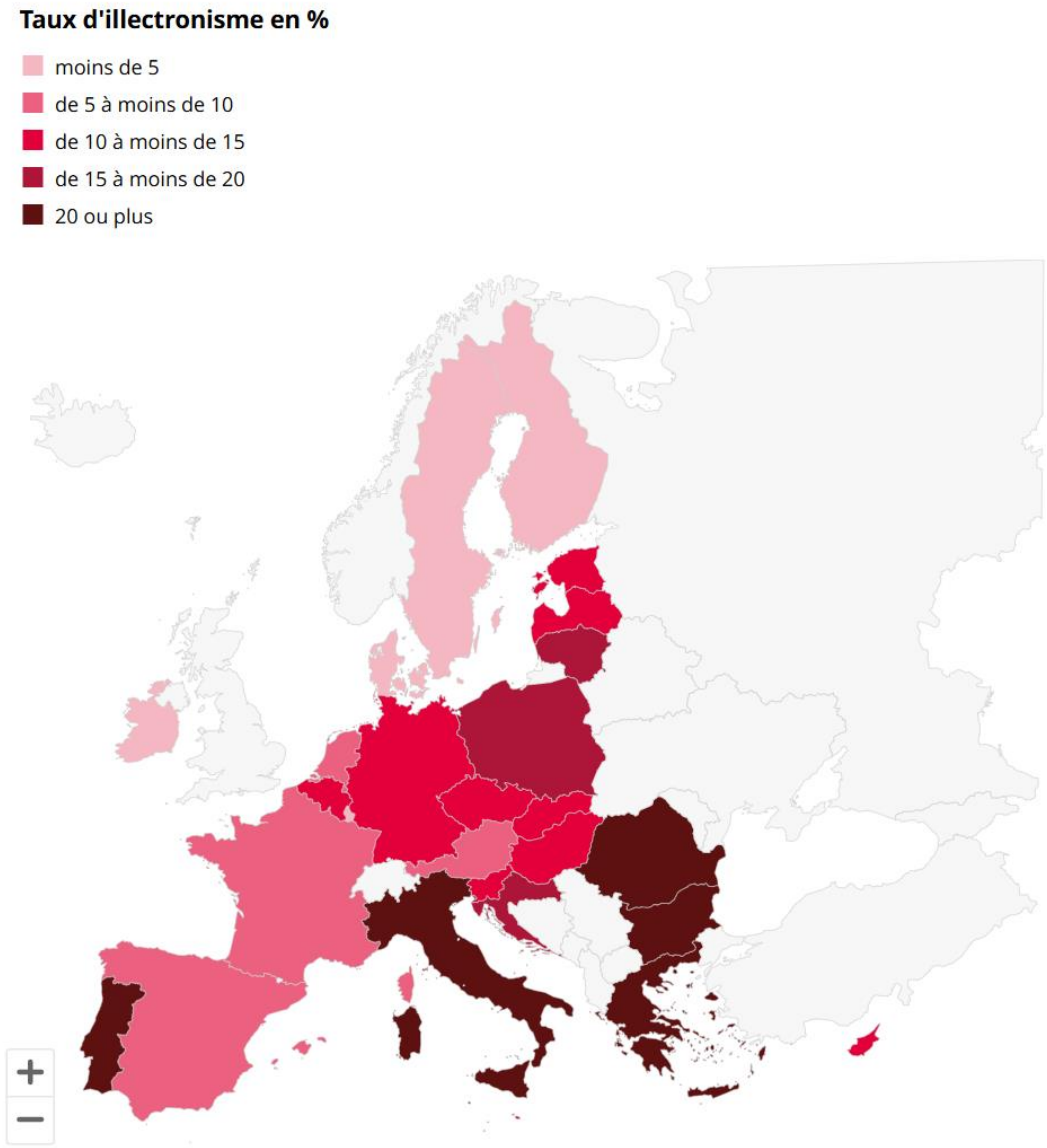
Source : Insee, enquête TIC ménages 2021.

https://www.insee.fr/fr/statistiques/7633654



In Europe

Part des personnes en situation d'illectronisme dans les pays européens



Lecture : en France, en 2021, 9,8 % des 16-74 ans sont en situation d'illectronisme.
Champ : Union européenne (UE27), personnes de 16 à 74 ans vivant en logement ordinaire.
Source : Eurostat, enquête TIC ménages 2021.

https://www.insee.fr/fr/statistiques/7633654



Is digital accessibility for the disabled?

TRUE / FALSE



Is digital accessibility for the disabled?

TRUE / FALSE

Digital accessibility is for anyone with a disability, permanent or temporary.

A person in a wheelchair has no trouble navigating but someone who has vision problems, color blindness, speaking difficulty, a broken arm...

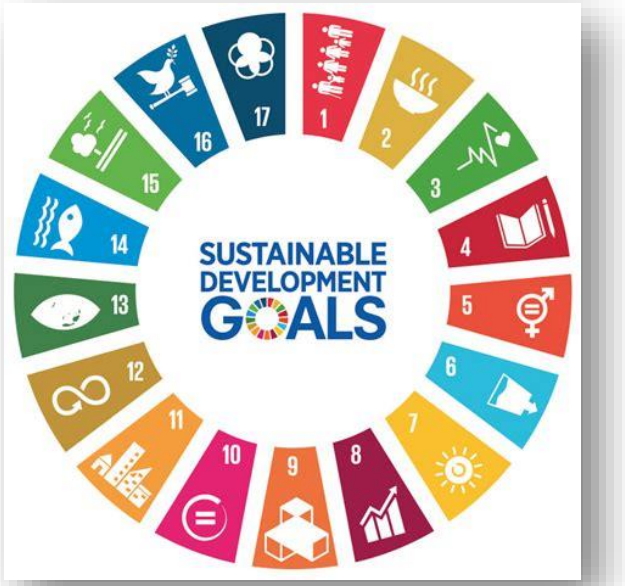
What is accessibility ?

Accessibility is the act of making a digital service accessible to **everyone** regardless of their **context of use.**

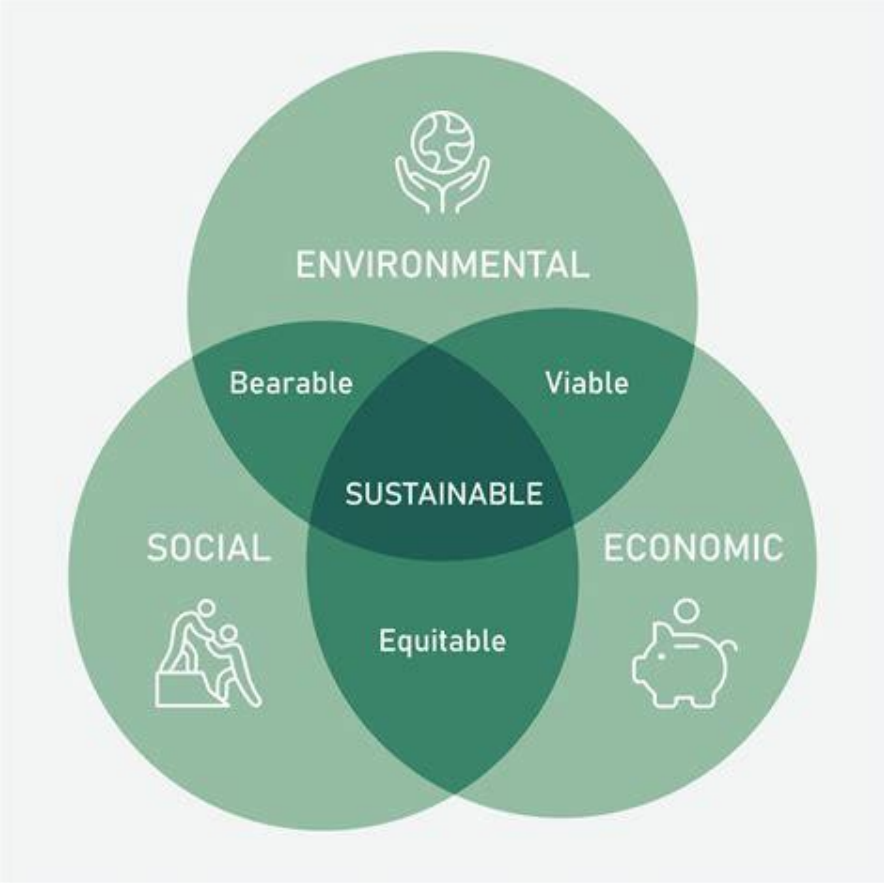
Responsible Digital

Responsible Digital

“Information and communication technologies whose economic, ecological, social and societal footprint has been **voluntarily reduced** and/or which help humanity to achieve the sustainable development goals.”



Sustainable development



3 commitments :

- Environmental Preservation (PLANET)
- Social Equity (PEOPLE)
- Economic Efficiency (PROFIT)

Staying below 1.5°C of global warming by 2100* means:

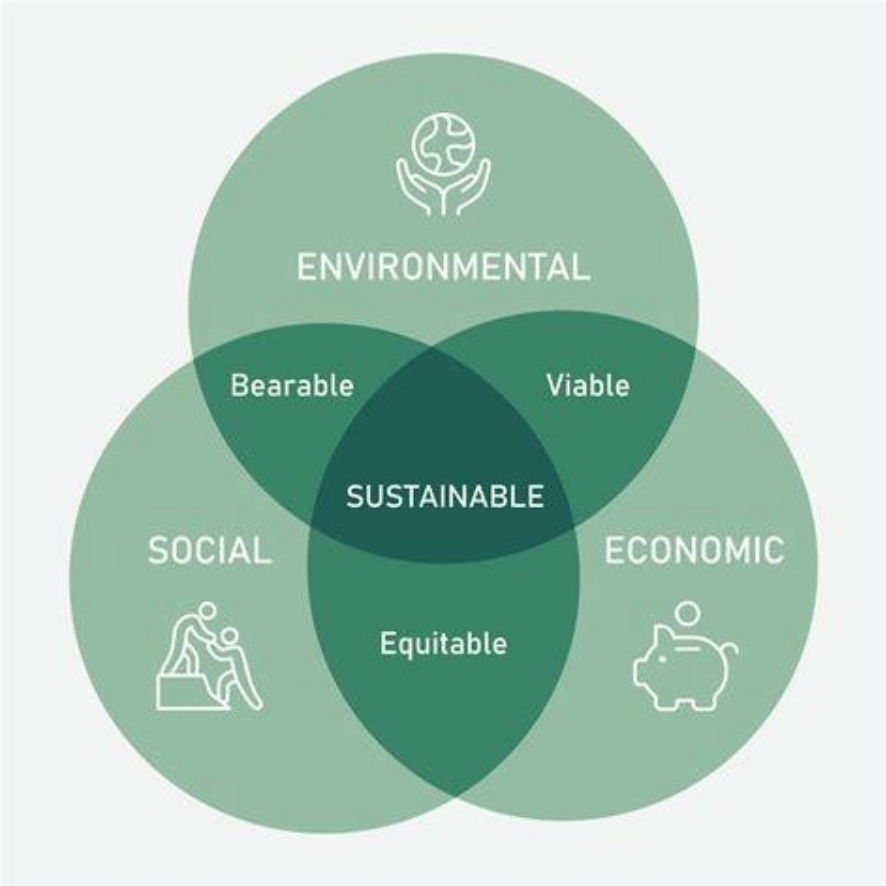
- **To divide by 2 our global emissions by 2030** (IPCC),
- Divide by 5 our emissions in France by 5 by 2050

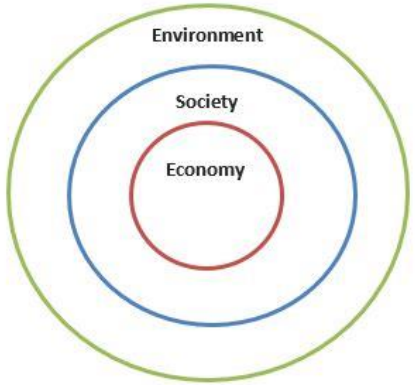
2 additionnal concepts
EFFICIENCY: Doing More with Less
EFFECTIVNESS: Doing Things Differently

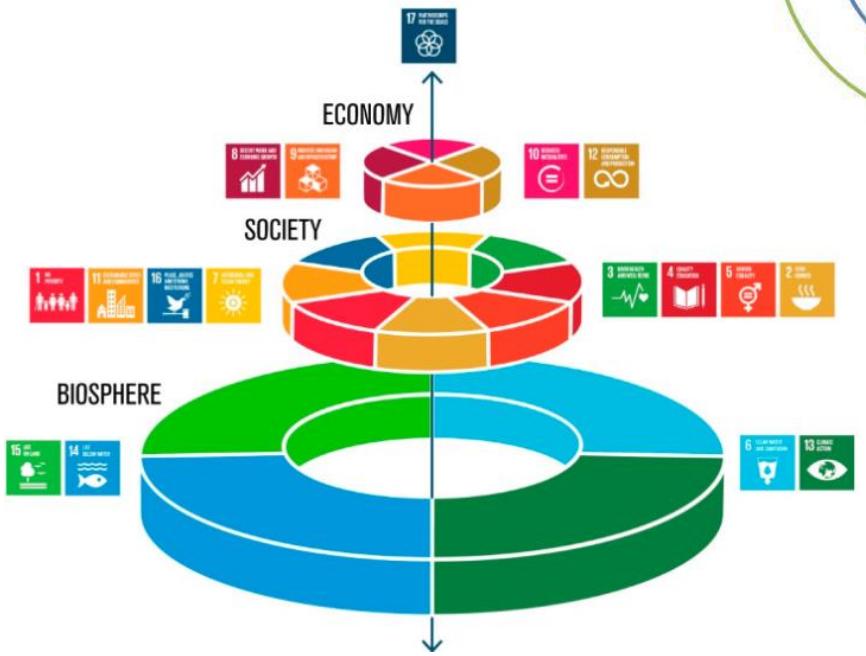
"Meeting the needs of present generations without compromising the ability of future generations to meet their own needs" Report by Mrs. Brundtland, 1987

*sources : * L'Accord de Paris, Nations Unies, Ministère de la transition écologique, Estimation de l'empreinte carbone de 1995 à 2020, octobre 2021*

A strong sustainable development

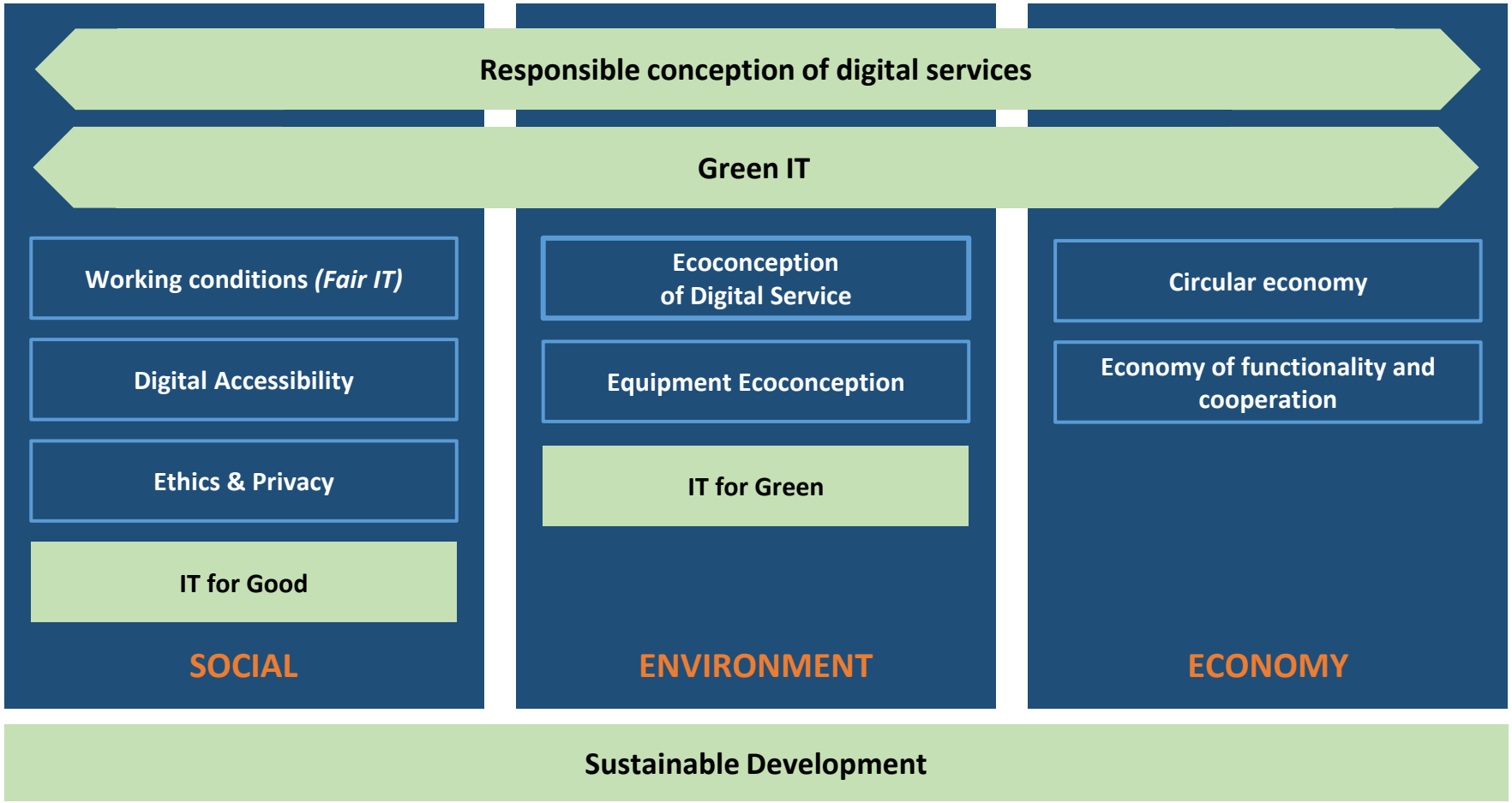






Example of an integration of the UN SDGs with a tripartite nested model of sustainability as proposed by Rockström and Sukhdev (2016).

Responsible Digital



sources : Du Green IT au numérique responsable : lexique des termes de référence, collectif Green IT, 2018,

Reduce environmental impact

Reduce the environmental footprint

during the design, use, and end-of-life of digital technology.

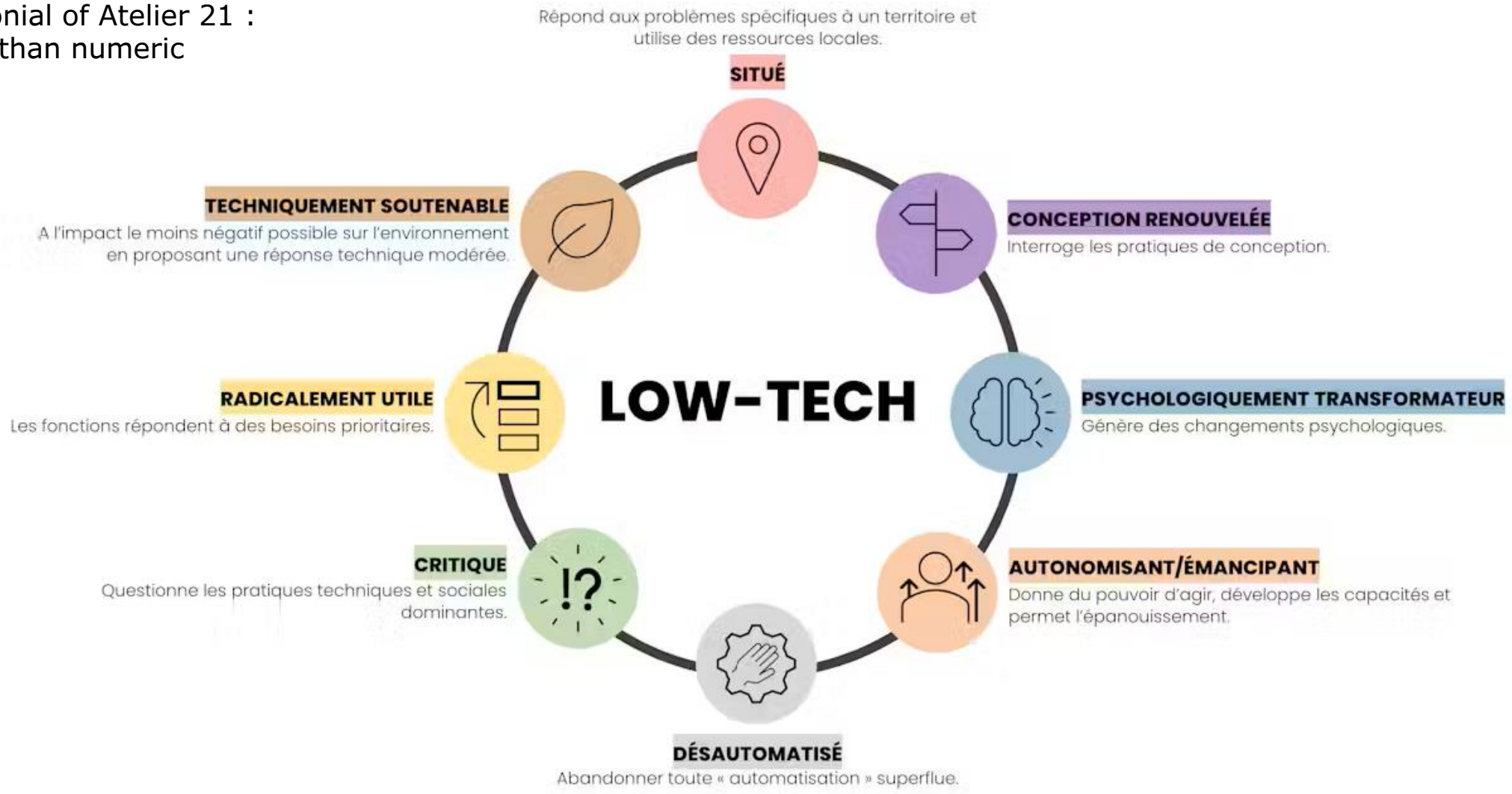
Green IT

Digital sobriety

Ecoconception of digital services

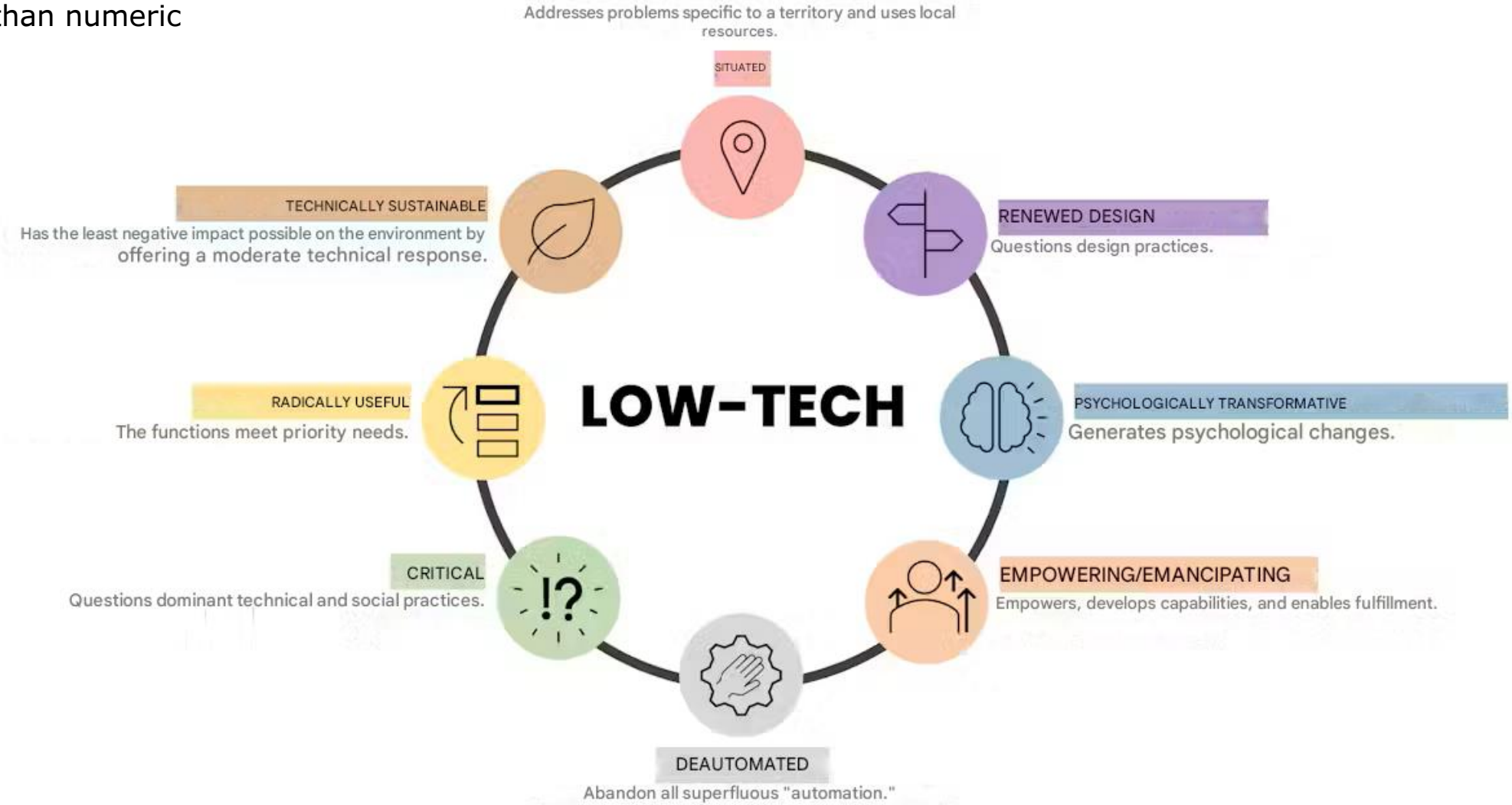
Low tech

- Testimonial of Atelier 21 : deeper than numeric



Adapté de : Martin, A., Colin, C. et Gaultier, A. (2022). Cartographie du concept low-tech : guider la conception vers des techniques soutenables. *Actes du 56ème Congrès de la SELF, Vulnérabilités et risques émergents : penser et agir ensemble pour transformer durablement*. Genève, 6 au 8 juillet 2022.

- Testimonial of Atelier 21 : deeper than numeric



Adapted from: Martin, A., Colin, C., and Gaultier, A. (2022). Low-Tech Concept Mapping: Guiding Design Toward Sustainable Techniques. Proceedings of the 56th SELF Congress, Emerging Vulnerabilities and Risks: Thinking and Acting Together for Sustainable Transformation. Geneva, July 6–8, 2022

Reduce the social and societal impact

Working Conditions during Manufacturing

Conditions of use for inclusion of everyone

Fair IT

Digital accessibility

Electronic illiteracy

Fair Cobalt Alliance
L'Alliance du Cobalt Equitable

FAIRPHONE

EMMAÜS CONNECT

Accompagner les plus isolés dans leurs premiers pas numériques

Digital to achieve SDG

Use of digital technology
- to reduce the environmental footprint of a service
- or to support a social/humanitarian action

IT for good

IT for green









Legislation Ecosystem

Legislations to support the changes



- **Energy Transition for Green Growth Act**

Planned obsolescence offense punishable by two years' imprisonment and a fine of €300,000 or 5% of turnover.

- **'AGEC' Law (Anti-Waste for a Circular Economy)**

Reparability Index (2022) and Durability Index (2024)
Spare parts for more than 5 years, Updates for 2 years
Public authorities must have 20% of their purchases reconditioned or made with recycled materials
Telecom operators publish the environmental impacts

- **Climate and Resilience Law**

Display of the environmental impact of goods and services, with a multi-criteria LCA

- **'REEN' Law (Reducing the Environmental Footprint of Digital Technology)**

Raising awareness of digital sobriety (primary-university)
Training in ecodesign and LCA of digital services (engineering qualification)
It is forbidden to restricte consumers' freedom to install the software or operating systems of their choice



A European Parliament resolution adopted on November 25, 2020, calls for Europe **to move toward a more sustainable single market** for business and consumers: "*establishing sustainable production and consumption strategies that **take into account planetary boundaries**, with priority given to more efficient and sustainable use of our resources.*"

- WEEE Directive - 2002-2012
- RoHS Directives – 2002, 2008, 2011
- REACH Regulation 2006
- Directive on Ecodesign Requirements for Energy-Related Products 2005-2009
- Battery Directive - 2006 and revision in 2022

Maximum quantities of certain chemical components and metals, Collection, recycling, and disposal obligations. Increased lithium recovery rate in batteries.

Ecosystem


ADEME
AGENCE DE LA TRANSITION ÉCOLOGIQUE


arcep


greenIT.fr


AGIT
ALLIANCE GREEN IT


THE SHIFT PROJECT
THE CARBON TRANSITION THINK TANK


Cigref
RÉUSSIR LE NUMÉRIQUE


RÉPUBLIQUE FRANÇAISE
Liberté
Égalité
Fraternité


mi
num_
eco


DesignGouv
Par le pôle design des services numériques


Inria


INR
Institut du Numérique Responsable




TCO
CERTIFIED

To conclude…

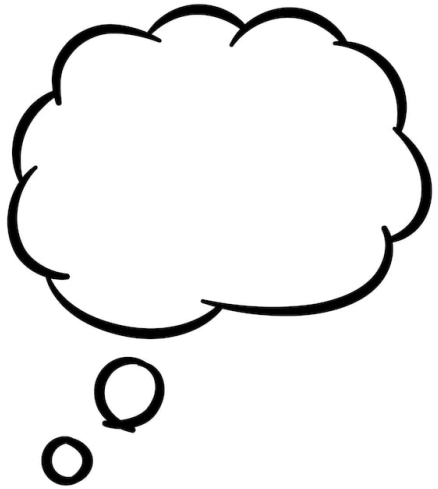
What are your feelings ?

Yes, I know but we cannot stop using digital tools

Yes, we have an issue

So what...

It is not my priority...

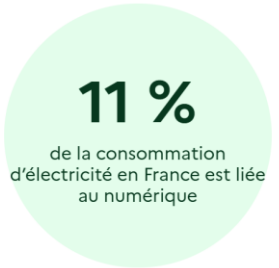


Too hard to work on

Lesson 1 - As a summary

- Digital is growing fast with intensity
- The main environmental impact is user equipment manufacturing
- IA accelerates datacenter intensity (size, number, energy)
- Digital is about 10 % of global electricity need.
- In addition of environmental impact, different social impacts *(labour conditions, ethics...)* are present during all lifecycle of digital services.

https://mtaterre.fr/articles/le-numerique-responsable-cest-quoi/





Planning

Groupe 1

DATE	HOUR	CONTENT
September 17	8.00	Main environmental and social challenges of digital Digital Responsible
	10.20	MOOC INR – Modules 1 - 3
September 24	8.00	The components of the Information System - Zoom / Datacenter & IA
	10.20	MOOC INR – Modules 4 - 11
October 1st	8.00	AI Battle (Association Latitudes)
	10.20	TP Devices diagnosis (Christophe Denoyelle)
October 8	8.00	Environmental assessment / Part 1: Life Cycle Assessment (LCA)
	10.20	MOOC INR – Modules 12 à 14 – Certification preparation
November 12	8.00	Environmental assessment / Part 2: LCA in relation to TP Diagnosis Digital accessibility
	10.20	Testimonial Alumni ISEN – Thomy Gensollen
November 19	10.20	Testimonial MC2i "Making digital transformation a positive revolution"
	13.30	Eco-design of digital services - TD eco design
November 26	8.00	TD presentation (evaluation) - Conclusion
	10.20	Testimonial Atelier 21 – Low Tech

JUNIA Grande école d'ingénieurs

HEI·ISEN·ISA

Septembre -Décembre 2025

ISEN
Module
Numérique Responsable